

Work Stream 6: Measure Theory, Genericity, and the Codimension of the Blow-up

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September 2026

1 Introduction

A profound discussion between mathematicians in the community ([u/Spidero0w0o](#), [u/cowgod42](#), [u/ComprehensiveWash958](#)) centered on how to rigorously describe the “unlikeliness” of the singularity using measure theory on infinite-dimensional Hilbert spaces ($H^s(\mathbb{R}^3)$).

2 Shyness and Prevalence

Because Lebesgue measure does not exist in infinite dimensions, we apply the Hunt–Sauer–Yorke definition of *shyness*. We hypothesize that the set of initial conditions (or forces) leading to the OpenAI blowup is a shy set.

An earlier draft supported this by the conditioning of the 5×5 linear system that matches the five radial moments of the inner and outer profiles (Proposition B.8 and Eqs. (5.10)–(5.11) of the OpenAI Navier–Stokes paper). The raw, dimensional matrix A has a condition number growing as $\kappa(A) \sim X_R^{7.75}$, reaching $\sim 10^{28}$ at $X_R = 1000$. This is a non-dimensionalization artifact: A mixes entries spanning X_R^0 to X_R^8 , and after diagonal scaling $A = DBD$ the dimensionless system has $\kappa(B) \approx 4.11 \times 10^5$, independent of X_R and numerically benign (see `OpenAI_NSE_Verification.tex`, §3). The condition number of a linear system used to *choose parameters in a proof* is in any case not a dynamical instability of the flow, and it says nothing about shyness. The measure-theoretic hypothesis therefore stands on its own and needs a different tool.

3 The Rigorous Version: Codimension

The precise meaning of “non-generic” for a self-similar blow-up is its *codimension*: linearize the leading-order self-similar profile in similarity variables and count the unstable eigenmodes beyond those generated by the symmetries of the equation (space and time translations, scaling). A profile with m genuinely unstable directions is reached from a codimension- m set of nearby data and forces; $m = 0$ is a stable, hence generic, blow-up. This is exactly the question that computer-assisted and neural-network methods have answered for Boussinesq and Euler profiles: Chen and Hou established a stable nearly self-similar blow-up with smooth data [1], and Wang et al. discovered families of *unstable* self-similar singularities with a definite number of unstable modes [2]. A finite, nonzero m for the OpenAI core would give “shy” a concrete meaning; it would *not* make the blow-up irrelevant, since unstable self-similar solutions still organize transient dynamics. Computing m for the linearized OpenAI profile is the open problem this work stream now proposes in place of the condition-number argument.

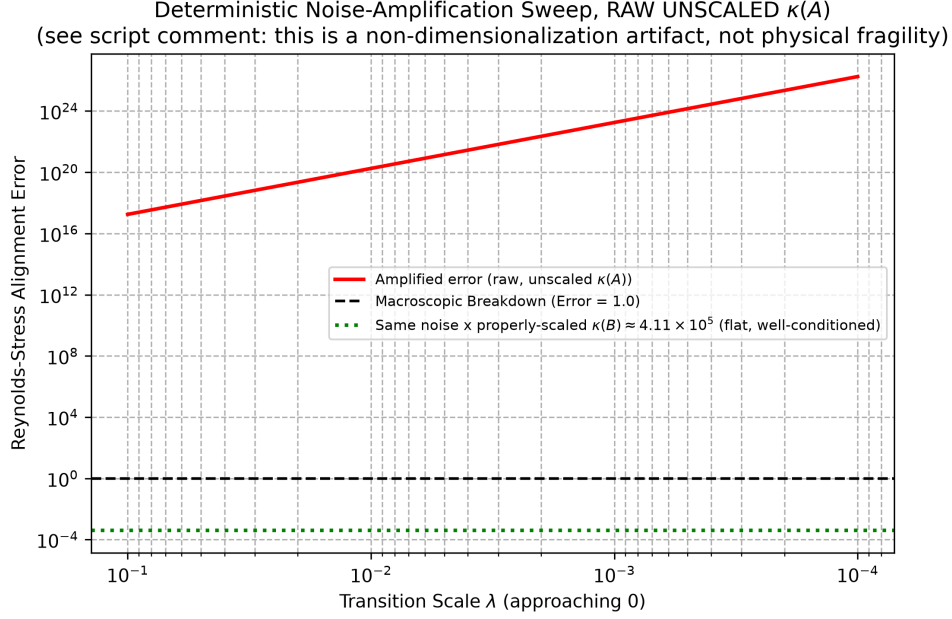


Figure 1: Deterministic noise-amplification sweep of the raw, unscaled moment-matching matrix A as a function of the transition scale λ . The apparent amplification of a 10^{-9} m/s perturbation to order one is a property of the unscaled coordinates; the properly non-dimensionalized system (green line, $\kappa(B) \approx 4.11 \times 10^5$) shows none of it. The figure is retained only to document why the earlier “ 10^{28} fine-tuning” reading was withdrawn.

References

- [1] J. Chen, T. Y. Hou, “Stable nearly self-similar blowup of the 2D Boussinesq and 3D Euler equations with smooth data,” arXiv:2210.07191, 2022.
- [2] Y. Wang et al., “Discovery of unstable singularities,” arXiv:2509.14185, 2025.

Acknowledgments

We thank the rigorous mathematical community on `r/math` and `r/FluidMechanics` for highlighting the need for infinite-dimensional measure theory.